Our Capstone Idea, Explained Simply: Crux-Audited Consensus

Oct 4, 2026 · @swaroop

The idea in one paragraph

When two strong AI models agree on an answer, the weaker "judge" model that is supposed to check them simply accepts the agreement, even when both are wrong. We fix this by making the two strong models name the one point their answer depends on (the **crux**), and say what their answer would be if that point were true or false. The weak judge then checks only that one point, which is much easier than checking the whole argument. We also give a way to tell, without knowing the right answer, whether a crux is worth checking, and we use that to train the strong models to produce good cruxes.

Background: why a weak checker has to check a strong AI

AI models are getting better than the people (or smaller models) who are supposed to check their work. If a model solves a hard physics problem, a non-expert cannot tell whether the answer is right. This problem is called **scalable oversight**: how can a weaker checker reliably supervise a stronger system?

It matters in practice, not only in theory. When companies train a new model, they often use an older, smaller model to grade its answers (this is called RLAIF, reinforcement learning from AI feedback). If the grader can be fooled, the new model learns to fool it instead of learning to be right.

An everyday analogy: a school principal has to decide which of two expert teachers is right about a hard exam question. The principal is not an expert in the subject. The question is how to set up the conversation so the principal still ends up with the right answer.

What the base paper did

Our base paper is Collaborative Disagreement Resolution for Scalable Oversight (Jiang et al., University of Chicago, published at ICML 2026). It compares two ways of letting a weak judge use two strong models.

- **Debate (the old way):** each strong model is stuck defending its own answer and tries to convince the judge. The judge must pick a winner. The problem: the model that sounds more convincing can win even when it is wrong.
- **Disagreement Resolution, or DR (their new way):** the two strong models act as collaborating "consultants". They compare their reasoning, find the point where they disagree (the crux), and either agree on one answer or hand the judge that one disputed point.

With small judges (GPT-4o-mini, Gemma-3-4B) and strong consultants (GPT-4o, Claude Sonnet 4, GLM-4.6, Kimi K2), DR raised the judge's accuracy to 62.1% on average, against 49.2% for debate, on hard question sets (GPQA, SuperGPQA, Humanity's Last Exam).

The problem we found

The weak judge in DR is not really checking anything. It is mostly saying "yes" to whatever the two consultants agree on. Three numbers from the paper's own results show this:

1. The two consultants reached agreement in **at least 96%** of cases.
2. When they agreed, the judge accepted their answer **at least 99.5%** of the time.
3. So the crux was handed to the judge to decide in **4% of cases or fewer**.

This works when the consultants agree on the right answer. But sometimes they talk each other into the **same wrong answer**. The paper calls this the **agreement trap**, and the judge has no way to catch it, because it just accepts the agreement.

The authors know this is a weakness. Their impact statement says agreement traps are a risk and asks future work to build safeguards against them. Their method also only uses prompting (no training), and assumes both consultants are honest.

Our fix, in plain words

Before the judge accepts an agreement, the consultants must hand over three things:

1. **The crux:** the one claim their answer depends on.
2. **Their resolution of it:** do they think the claim is true or false?
3. **Two "if" answers:** what their final answer would be if the claim were true, and if it were false.

The judge checks only the crux. The final answer is whichever "if" answer matches the judge's verdict. So even if both consultants agreed on the wrong answer, a correct check of the crux brings back the right one.

A crux is only worth checking if it has two properties, and both can be measured without knowing the right answer:

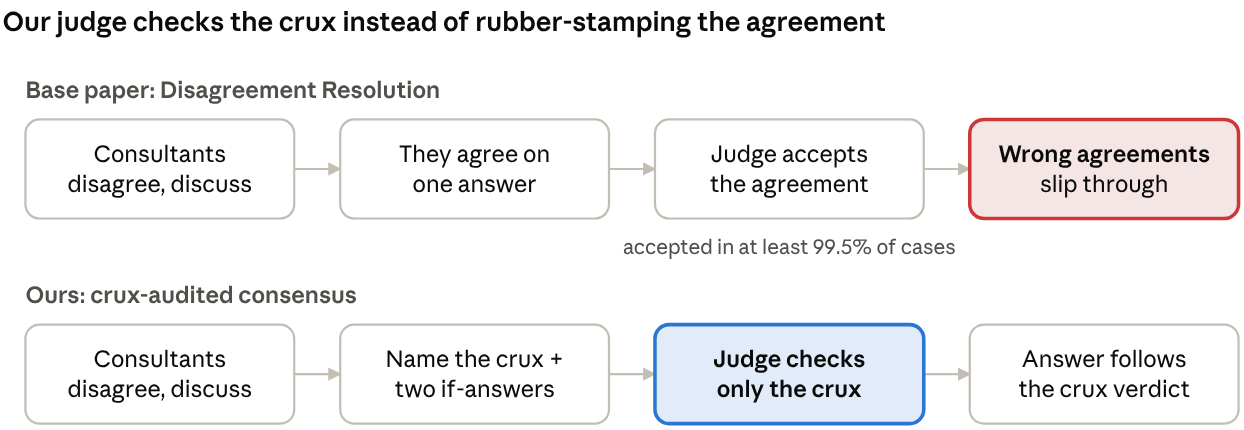
- **Decisive:** settling the claim settles the answer. We test this by asking each consultant the two "if" questions; if their answers flip with the claim, it is decisive.
- **Checkable:** the weak judge is more reliable on the claim than on the original question. We test this by asking the judge the same thing several times and seeing how consistent it is.

Worked example. Question: is 91 a prime number? (Right answer: no.)

- Consultant A says "prime"; consultant B says "not prime, because 7 × 13 = 91". After discussing, B gets talked round and both say "prime". That is an agreement trap, and in DR the judge would accept it.
- In our protocol, they must name the crux: "7 × 13 = 91". If true, the answer is "not prime"; if false, "prime". It is decisive.
- A small judge cannot easily test whether 91 is prime, but it can multiply 7 × 13. It is checkable.
- The judge finds the claim true, so the final answer is "not prime". The trap is caught.

A bad crux would be "91 is an odd number": easy to check, but it does not decide anything. Our score gives it zero, which is why two models that simply agree cannot earn reward with a fake crux.

The two flows side by side



base paper vs our protocol · 4 steps each

The first two steps are the same. The difference is what happens after the consultants agree: the base paper's judge accepts the agreement, while ours checks the one claim it rests on.

How we will test it

We use free, open models that run on college GPUs, so no paid APIs are needed:

- **Consultants (the strong models):** two 8-billion-parameter models from different companies, for example Qwen3-8B and Llama-3.1-8B.
- **Judge (the weak model):** a much smaller model, for example Qwen2.5-1.5B.
- **Questions:** multiple-choice and maths sets with known answers (MMLU-Pro without the health topics, GPQA Diamond, hard MATH problems). We keep only questions where the two consultants disagree and one of them is right, as the base paper did.

The right answers are used only to score the experiment, never inside the method.

The kill test (first ~3 weeks, cheap): for about 300 questions, we run DR, pull out each crux, and compare two numbers: how often the judge is right about the crux, and how often it is right about the full question. If the judge is not clearly better on cruxes, the core idea does not work and we switch to our backup topic early, before spending the year on it.

What we get if it works, and if it fails

Outcome	The paper we write	Realistic venue
It works	Crux-audited consensus beats DR, mainly by catching agreement traps; a method to score cruxes without labels	NeurIPS 2027 (Paper 1)
Training also works	Consultants trained on our crux score give better cruxes than ones trained to please the judge	ICLR 2028 (Paper 2)
Main idea fails	An analysis showing DR's gains come mostly from the judge rubber-stamping agreement, plus a scored set of cruxes others can reuse	Workshop or ACL/EMNLP Findings

Even the failure case gives a publishable result, because DR is a fresh ICML 2026 paper and nobody has measured how good its cruxes actually are.

What already exists, and what is new

Paper	What it does	What it does not do
Disagreement Resolution (ICML 2026)	Consultants collaborate and find a crux	Never checks the crux behind an agreement; prompting only
ColMAD (2025–26)	Collaborative debate, framed as a game	Skips the judge when agents agree; prompting only
Prover-Estimator Debate (theory)	Proves how to reward picking the right sub-claim	No experiments with real language models
CruxBench (Sep 2026)	Scores cruxes by how much they shift belief, for forecasting	Not about two agents or a judge; no training
Peer prediction (ICLR 2026)	Rewards honest answers without labels	Rewards answers, not cruxes

What is new in ours: the judge checks the crux behind every agreement; a label-free score for whether a crux is decisive and checkable; and training the consultants on that score.

We searched for follow-up work on DR on 4 October 2026 and found none that adds crux checking or training.

Questions BJD may ask

Is this agentic AI? It is multi-agent LLM work: two agents interact and a third supervises them. It is framed as AI oversight rather than tool-using agents, so we should ask him directly whether this counts. If needed, Phase 4 can move to coding agents, where a crux is checked by running a test.

Isn't this just DR plus a reward (method A + method B)? No. The new insight is about what makes a crux useful to a weak judge: it must be decisive and checkable, and both can be measured without labels. That changes the protocol (the judge checks the crux behind every agreement) and gives a reward that faked agreement cannot earn. Neither exists in DR or its neighbours.

Why can't the judge just answer the question itself? It is too weak: in DR's experiments the judges alone scored 11–26% on the disputed questions. The point is that a hard question often hinges on one easier claim.

What if a consultant lies on purpose? DR assumes honest consultants. We test this in Phase 4 with one deliberately dishonest consultant. Prover-estimator debate gives the theory for why honest play can still win.

What if it doesn't work? We find out in about three weeks with the kill test. Even then, the analysis of DR's cruxes is a publishable result, and we have a backup topic ready.

Do we need paid APIs or big compute? No. Everything uses open 1.5B–12B models on college GPUs. Training (Phase 4) starts with offline preference training (DPO), which is far cheaper than full reinforcement learning.

Glossary

Term	Meaning
Scalable oversight	Getting a weaker checker (a person or small model) to reliably supervise a stronger AI
Consultant	One of the two strong models that answer the question and discuss it
Judge	The weak model that makes the final call
Crux	The single claim that the disagreement, or the agreement, depends on
Agreement trap	Both consultants end up agreeing on the same wrong answer
Decisive crux	Settling the claim settles the final answer
Checkable crux	The weak judge is more reliable on the claim than on the full question
Label-free	Measured without knowing the correct answer
RLAIF	Training a model using grades from another AI model
DPO	A cheap way to train a model from pairs of better and worse outputs
Kill test	An early, cheap experiment that tells us whether to drop the idea